Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/31/2024
PI Name:	Crowe, Kelly Ph.D.		
Project Title:	Assessment of Sialylation in Skeletal Muscle Atrophy Due to Simulated Microgravity		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Food and Nutrition**:Risk of Performance Decrement and Crew Illness Due to Inadequate Food and Nutrition (2) **Muscle**:Risk of Impaired Performance Due to Reduced Muscle Size, Strength and Endurance		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	45207-1035	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2023 HERO NNJ23ZSA001N-OMNIBUS : NASA Human Research Program Omnibus Opportunity
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2025 per NSSC information (Ed., 11/27/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mortreux, Marie Ph.D. (University of Rhode Island)		
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Task Description:	Long-term skeletal muscle unloading due to microgravity exposure in spaceflight conditions has a striking effect on the health of skeletal muscle, which may impact performance of flight mission tasks. While resistance exercise has shown efficacy as a countermeasure, capacity constraints may preclude effective implementation of these exercise protocols on the lunar surface. As such, it is crucial to identify novel targets to ameliorate skeletal muscle atrophy. One such target is sialic acid (SA), a terminal glycan on extracellular glycan chains, as changes in SA are associated with skeletal muscle weakness and wasting due to their role in development, regeneration, and contractility. In the rare muscle disease GNE myopathy, a reduction in SA levels leads to progressive muscle weakness and wasting. Furthermore, SA alterations have been seen in atrophy; for example, expression of sialidases, which remove SA from glycoproteins, have shown alterations in a model of atrophy. While global perturbations in SA levels certainly impair skeletal muscle function, the linkage of SA appears to mediate these effects; in skeletal muscle of patients with GNE myopathy, hyposialylation of O-linked, but not N-linked glycans, was correlated to pathology. We hypothesize that simulated microgravity will lead to linkage-specific alterations in skeletal muscle sialylation mediated by changes in sialidases and/or sialyltransferases. To address this, we will use immunofluorescence with a panel of SA-detecting lectins, glycanbinding proteins, as well as lectin microarrays, to probe alterations in linkage-specific SA abundance in skeletal muscle tissue from rats with or without a hindlimb-unloading protocol. Next, we will probe expression of the sialyltransferases that form SA linkages and the sialidases that enzymatically remove them, via qRT-PCR and Western blotting to assess the underlying mechanism of these SA alterations and identify novel pharmacological targets. This will be the first study to address alterations in skeletal muscle sialylation in the context of spaceflight-induced muscle atrophy. This would inform future studies to assess rescue of sialylation in skeletal muscle as a treatment for atrophy, as skeletal muscle sialylation can be altered via oral supplementation and gene therapy; in fact, sialylation-altering nutraceutical therapies for skeletal muscle disease have recently progressed through clinical trials for the rare disease GNE myopathy. This work aims to develop novel targets to address skeletal muscle atrophy via nutraceutical and pharmacological approaches, which would help to ensure effective performance of flight mission tasks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This will be the first study to address alterations in skeletal muscle sialylation in the context of spaceflight-induced muscle atrophy. This would inform future studies to assess rescue of sialylation in skeletal muscle as a treatment for atrophy, as skeletal muscle sialylation can be altered via oral supplementation and gene therapy; in fact, several glycan-altering therapies for skeletal muscle disease have recently progressed through clinical trials. This work represents a novel target to address spaceflight-relevant skeletal muscle atrophy, which could potentially help to inform treatment for other types of skeletal muscle atrophy such as sarcopenia and cachexia.
	Accomplishments and Progress Towards Aims: Specific Aim 1: Determine linkage-specific skeletal muscle sialylation in a rodent hindlimb-suspension model to simulate microgravity exposure. The goal of this aim is to use both lectin staining and lectin microarray to look at changes in sialic acid abundance in both male and female rats with and without a hindlimb-unloading protocol for 14 days, to induce unloading atrophy. Both gastrocnemius (a predominantly fast-twitch muscle) and soleus (a predominantly slow-twitch muscle) were used for further analysis. Progress is ongoing for both methods as described below: Lectin Staining: We have successfully stained the gastrocnemius and soleus for all 14 hindlimb- unloading and 14 control rats with several lectins. We used sialidase, an enzyme that removes sialic acid, as a control for sialylation-detecting lectins; only those lectins in which we were able to distinguish between sialidase treated and untreated samples were used for further analysis. Based on this criterion, we collected images using time-matched fluorescent microscopy for four lectins: Wheat Germ Agglutinin (WGA, recognizing any bound SA), Sambucus nigra agglutinin (SNA, recognizing Siaa2-6Gal/GalNAc), and Maackia amurensis agglutinin (MAA, recognizing Siaa2-3Gal). These images are in the process of being analyzed using ImageJ to determine relative abundance of lectin binding. Data on total relative fluorescence has been collected for all images, and data on fiber diameter quantification (a key normalization) is underway. Sample images from each lectin stain are shown below. As quantification is still ongoing, we are still blinded as to sample identity; when data collection is finalized, we will unblind, and collected data will be analyzed as relative lectin fluorescence normalized to fiber size. Lectin Microarray: Upon further interaction with the vendor quoted in our grant application, we were concerned that they did not perform this particular assay routinely (and thus would not be able to provide us with sufficiently detailed sample preparation protocols). To remedy this, we found an academic collaborator who more routinely performs this application and would be able to perform the lectin microarray with more confidence and at a reduced cost. The sample preparation protocol sent from this collaborator’s lab necessitates subcellular fractionation protocol to remove the cytoplasmic fraction, as many of our target proteins and glycans of interest are membrane bound or contained in the Golgi apparatus. This requires use of a specialty adapter for our centrifuge, which has been ordered; we anticipate a trial run of this using practice samples within the next few weeks. Based on limited availability of project sample, we will be using this collaborator’s extraction protocol on a limited number of test samples and sending them to her laboratory for testing before preparing our limited project samples using this protocol. Once the methodology has been validated with test samples, we will perform the sample preparation procedure on our project samples and send them to the University of Alberta for lectin microarray analysis. Specific Aim 2: Assess expression of sialyltransferases and sialidases in skeletal muscle of a rodent hindlimb-suspension model. The goal of this aim is to use quantitative reverse transcription polymerase chain reaction (qRTPCR) and Western Blotting to probe changes in sialidases and sialyltransferases, enzymes that could be responsible for altering sialic acid abundance in both male and female rats, with and without a hindlimb-unloading protocol for 14 days, to induce unloading atrophy. Again, both gastrocnemius and soleus were used. Progress for both methods is ongoing as described below:

Task Progress:	qRTPCR: While we had initially planned to use a singleplex approach, in an effort to preserve sample we have decided to instead employ a duplex method. In this method, we use a Taq DNA polymerase-based assay with VIC and fluorescein (FAM) fluorescent reporter dyes for the endogenous control and gene of interest respectively. This allows for simultaneous quantification of our gene target and housekeeping gene in a single well, which halves the amount of sample needed to quantify all of our targets. This is particularly important for the soleus muscle, which is quite limited. After trialing both GAPDH and 18s ribosomal RNA as endogenous control genes, we selected 18s. We have validated our duplex approach for 4 targets; Neu1, Neu3, St3gal3, and St8sia2 (two sialytransferases and two sialydases) in rat skeletal muscle samples by demonstrating that the amplification of both the 18s endogenous control and our genes of interest is acceptably similar when run in singleplex and duplex. We are in the process of validating our duplex approach for each primer set and probing gene expression in target tissues. Western Blotting: Given the relatively small size of the rat soleus muscle, we first sought to maximize the yield and versatility of each protein extraction done using test samples. We therefore worked to optimize our protein extraction methods to achieve the greatest mg protein/mg tissue yield. We trialed a series of sonication, incubation, and buffer conditions and quantified protein content using bicinchoninic acid (BCA). In a trial using practice tissue, we achieved consistent yields averaging 27 ug protein/mg tissue in a buffer that is appropriate for our assays. This yield should allow us to complete our proposed experiments with the limited sample available. We next sought to validate our antibodies for Western blotting using test samples. In validating the sialidase and sialyltransferase antibodies with test samples prior to use of our limited project samples, we have developed unexpected concerns about the ability of commercially available antibodies to distinguish between various sialyltransferases (e.g., ST3GAL1, which catalyzes the addition of a2,3-linked sialic acids, and ST6GAL1, which catalyzes the addition of a2,6-linked sialic acids). This is due to unexpected similarities between these blots with antibodies for these enzymes with biologically distinct functions. As such, we are investigating use of mass spectrometry to probe protein alterations in these tissues. We have identified an academic partner who has significant experience in protein analysis of tissues with mass spectrometry, including in skeletal muscle tissues after hindlimb unloading. Again, we are looking to validate our methods with test samples before using limited project tissues. As such, we are sending sample tissues to this collaborator (these will be prepared in the same manner as the lectin microarray test samples mentioned above) to validate the protocol. Afterwards, we will prepare and send this collaborator our project tissues. How the results have been disseminated: As it was near the beginning of the granting period, preliminary data and grant plans were presented at the 2024 NASA Human Research Program (HRP) Investigators' Workshop (IWS). Initial data collected during the granting period will be presented at the American Society for Gravitational and Space Research (ASGSR) 2024 meeting and the 2025 HRP IWS. As project data is finalized, it will be submitted to a peer reviewed journal and appropriately deposited in a NASA data repository. [Note: See also the Cumulative Bibliography (Ed., 11/24/24).] Presented and/or abstract submitted: Mackey, B., Mazza, N., Cullen, R., Mortreux, M., & Crowe, K.E. Skeletal Muscle Sialylation in Simulated Microgravity. Poster presentation delivered at the Human Research Program Investigators' Workshop (HRP IWS), February 2024. Green, T., Greulach, E., Hornberger, C., Leontescu, N., Mathis, J. Parker, O., Simone, E., Vonderhaar, R., Woods, J., Mortreux, M., Wetzel, H., Crowe, K. E. Characterization of Skeletal Muscle Sialylation in a Ground-Based Rat Model of Microgravity. Abstract submitted for American Society for Gravitational and Space Research's (ASGSR) Annual Meeting, December 2024. Greulach, E., Hornberger, C., Leontescu, N., Mathis, J. Parker, O., Simone, E., Vonderhaar, R., Woods, J., Mortreux, M., Wetzel, H., Crowe, K. E. Assessment of sialylation in skeletal muscle of a ground-based rat model of microgravity. Abstract submitted to the Human Research Program Investigators' Workshop (HRP IWS), January 2025. Future plans: As noted, the most pressing future plan is to validate sample preparation using ultracentrifugation, which is expected to be completed in the coming weeks as our adapter arrives. From there, test samples will be sent to our collaborators to validate this sample preparation method for lectin microarray and mass spectrometry, respectively. Once validated, project samples will be prepared and sent out for lectin microarray (Aim 1) and mass spectrometry (Aim 2) analysis. As noted above, the use of mass spectrometry-based proteomics is a change from the initial plans of Western blotting due to concerns with cross-reactivity of commercially available antibodies for our targets, which we noted during antibody validation. For lectin staining (Aim 1), we plan to complete image analysis of fiber diameter using Cellpose and ImageJ in a blinded fashion. While we have begun with a subset of lectins, we will likely expand the number of lectins used for lectin staining based on results from lectin microarray for validation purposes. For qRTPCR (Aim 2), we plan to continue validation of our duplex method and quantification of our project samples using this method. The primers that we have validated thus far have been appropriate for duplexing. Based on the timeline of validating the sample preparation for the lectin microarray and mass spectrometry, we are likely to request a no-cost extension prior to the end of the granting period.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Mackey B, Mazza N, Cullen R, Mortreux M, Crowe KE. "Skeletal muscle sialylation in simulated microgravity." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024

Abstracts for Journals and Proceedings	Green T, Greulach E, Hornberger C, Leontescu N, Mathis J, Parker O, Simone E, Vonderhaar R, Woods J, Mortreux M, Wetzel H, Crowe KE. "Characterization of skeletal muscle sialylation in a ground-based rat model of microgravity." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Abstracts for Journals and Proceedings	Greulach E, Hornberger C, Leontescu N, Mathis J, Parker O, Simone E, Vonderhaar R, Woods J, Mortreux M, Wetzel H, Crowe KE. "Assessment of sialylation in skeletal muscle of a ground-based rat model of microgravity." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025